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SURVEY

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A comprehensive survey on machine learning for workplace injury analysis: risk prediction, return to work strategies, and demographic insights

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Abstract

This survey paper explores the application of machine learning (ML) techniques in the domain of workplace injuries, focusing on three key areas: risk prediction, return to work (RTW) strategies, and demographic analysis. Through an extensive review of literature from January 2015 to July 2024, this paper examines the latest advancements in ML-driven approaches to workplace safety and identifies important research gaps. This paper highlights how classical ML techniques, such as ensemble models and decision trees, have become essential tools for identifying workplace injury risks, enabling more accurate interventions. It emphasizes the importance of leveraging ML in personalized RTW programs, which use data-driven insights to improve recovery outcomes and reduce economic demands. In the context of demographic analysis, this paper explores how ML algorithms can uncover disparities in injury rates across various age groups, industries, and occupations, underscoring the need for targeted safety measures. Moreover, research gaps are identified, particularly regarding the emerging potential of advanced ML techniques, such as deep learning and large language models (LLMs), for analyzing structured and unstructured safety data, methods that have not yet been widely applied in workplace injury research. As such, future research should apply recent advances in ML, integrating these approaches with comprehensive and accessible datasets to enhance the prediction and prevention of workplace injuries, provide more detailed analytics and insights, and improve safety protocols across all industries. This comprehensive survey is an invaluable resource for researchers and practitioners leveraging ML to address complex challenges in workplace safety.

Keywords: Workplace injuries and illness, Return to work, Machine learning

Introduction

In the field of occupational health and safety, understanding and mitigating workplace injuries are a significant concern for both employers and employees. This survey paper explores three critical areas of research that are instrumental in advancing our knowledge and strategies to address workplace injuries: the prediction and identification of risk factors, RTW or return to original work (RTOW) processes and

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predictors, and the analysis of injury rates, types, and consequences across different demographics and industries [1–3]. Previous studies have recognized the importance of these dimensions in workplace injury research, particularly with the integration of machine learning techniques for risk assessment, injury prediction, and return to work outcomes [1–5]. The focus of our review is on the applications of machine learning and other advanced analytics within this domain. To the best of our knowledge, this is the first survey paper related to workplace injuries focusing on the use of machine learning.

Identifying and predicting risk factors for workplace injuries is fundamental for effective proactive occupational health management. Using advanced machine learning techniques and predictive analytics, researchers seek to accurately determine the specific conditions and behaviors that are most likely to result in workplace injuries. This approach allows for the implementation of targeted interventions before incidents occur, significantly enhancing workplace safety. This research directly contributes to reducing the incidence and severity of workplace injuries. According to the US Bureau of Labor Statistics, the total cost of work injuries in 2022 was an astonishing $167 billion [6], emphasizing the significant financial toll these incidents take on businesses and society. This figure includes wage and productivity losses, medical expenses, and administrative costs.

In addition to preventing injuries, understanding the RTW processes and predictors is important in developing effective rehabilitation and reintegration strategies for injured workers. Successful RTW programs not only facilitate the recovery and reintegration of injured employees, but also mitigate the long-term financial impacts on both workers and employers [7–9]. Research in this area focuses on identifying the factors that influence successful RTW, such as the nature and severity of the injury, the type of work performed, and the support systems in place. Effective RTW programs are needed for maintaining workforce productivity and reducing turnover costs. The importance of this research is underscored by data from the National Safety Council (NSC) where the average cost per medically consulted work injury in 2022 was $40,000 [6]. This figure reflects the direct costs associated with medical treatment and lost wages, as well as the indirect costs related to reduced productivity and employee replacement.

The analysis of injury rates, types, and consequences across different demographics and industries provides valuable insights into the specific vulnerabilities and risks faced by various worker populations [10, 11]. This research is critical for developing targeted safety measures and policies that address the unique needs of different sectors and demographic groups. For instance, industries such as construction, manufacturing, and healthcare are known to have higher rates of workplace injuries, entailing tailored interventions to mitigate these risks [12–14]. Furthermore, demographic factors such as age, gender, and socioeconomic status can influence the likelihood and impact of workplace injuries. According to the US Bureau of Labor Statistics, transportation and material moving occupations recorded the highest number of days away from work due to injuries, with an incidence rate of 410 cases per 10,000 full-time equivalent (FTE) workers in 2021–2022 [15]. This statistic indicates the substantial risks that workers are exposed to in specific industries and the need for focused safety initiatives.

Our research aims to provide an comprehensive resource for those studying and addressing workplace injuries by focusing on predicting and identifying risk factors, understanding RTW processes and predictors, and analyzing injury rates, types, and impacts across diverse demographics and industries. What sets this survey apart is its comprehensive approach that not only reviews the latest advancements from January 2015 to July 2024 but also addresses critical gaps in the existing literature. We selected this timeframe for several reasons: (1) this period captures the significant advancements in machine learning technologies, particularly the rise of deep learning and its applications in different domains; (2) this timeframe coincides with the growing availability of large-scale datasets in occupational health and safety, which are crucial for effective ML applications; and (3) by extending our review to July 2024, we ensure that the most recent developments and applications in the field are captured, providing an up-to-date analysis of the state of the art. To assess as many works as possible, our search methodology within this timeframe included key terms like "work-related or workplace injuries", "return-to-work statistics", "workers compensation after workplace injury" and "occupational injuries return time" queried across multiple academic platforms such as Google Scholar, ResearchGate, ScienceDirect, and Springer. Moreover, unlike previous works such as the survey by Cancelliere et al. [4], our research covers the use of machine learning and new methodologies, such as integrating deep learning and large language models (LLMs) for analyzing unstructured safety data, which have been underutilized in the field. Additionally, techniques for anomaly detection (Isolation Forest, Autoencoders), ensemble modeling (Gradient Boosting, XGBoost, AdaBoost, CatBoost), and explainability (SHAP) are also explored in workplace injury analysis. By blending these innovative approaches, this paper offers significant contributions that are vital for advancing predictive models and RTW strategies, making it a vital resource for both academic research and practical applications in occupational health and safety.

This survey paper provides a comprehensive review of machine learning applications in workplace injury analysis. We begin by briefly discussing related survey works further supporting the uniqueness of our work. We continue by presenting a description and analysis of the reviewed literature. This is followed by a detailed examination of current ML applications in risk prediction, return-to-work strategies, and addressing demographic disparities in workplace injuries. We then discuss the key findings of our survey, highlighting emerging trends, and identifying research gaps. The paper concludes with a summary of the current state of the field and offers directions for future research.

Related survey work and comparative literature

The intersection of workplace injury prediction and return-to-work (RTW) analysis has received significant attention in recent years, with various approaches and methodologies being explored. This section presents a comprehensive review of related survey works, highlighting the novelty and contributions of our research.

Recent reviews have demonstrated the growing adoption of artificial intelligence in occupational health settings. Chaudhry and Choudhury conducted a comprehensive review of clinical AI applications in occupational health, which, while examining workplace injuries, maintained a broader focus on clinical applications [16]. Similarly, Maheronnaghsh et al. examined machine learning applications in occupational safety and

health, but their work addressed general workplace hazards without specifically focusing on injury events or RTW outcomes [17]. In the construction sector, Cavalcanti et al. presented a systematic review of machine learning approaches for accident prevention [18]. Although their work demonstrated the potential of ML in injury prediction, it was limited to the construction domain and did not address RTW or demographic factors, which are crucial components of our investigation.

The RTW literature has evolved significantly, with recent work incorporating machine learning approaches. Escorpizo et al. conducted a scoping review that specifically examined machine learning applications in RTW studies, highlighting both strengths and limitations of current approaches [19]. However, their review did not address the prediction of injury risk or demographic analysis, which are key components of our integrated approach. Earlier work, such as Eib et al., focused on sustainable RTW from a behavioral and social perspective, while Schultz et al. provided theoretical frameworks for RTW in musculoskeletal disorders [20, 21]. Although these works offer valuable insights into RTW processes, they lack the analytical and predictive components that characterize our research.

Recent literature has also explored various aspects of injury analysis. Dodoo and Al-Samarraie conducted a systematic review of factors leading to occupational injuries and fatalities, providing valuable insights into injury causation but without incorporating machine learning or RTW analysis [22]. Khairuddin et al. focused on text-based analytics for predicting injury causal factors, demonstrating the potential of NLP in accident analysis but limiting their scope to text-based approaches [23].

In summary, our work addresses several key gaps identified in the existing literature by providing an integrated approach that combines injury prediction and RTW analysis, which have traditionally been examined separately in previous studies. We extend beyond traditional text-based analytics by incorporating multiple data modalities and demographic factors, while our approach uniquely combines the analytical rigor of machine learning with comprehensive consideration of social and organizational factors. This integrated approach, which combines injury prediction with RTW analysis while considering demographic and organizational factors, represents a novel contribution to the field of occupational health and safety.

Surveyed papers

This section presents a detailed overview of the surveyed works. We organize the surveyed papers into three key categories: Risk Prediction, Return to Work Strategies, and Demographic Analysis. Each category is further divided into relevant subsections, and the papers are listed chronologically in ascending order within each subsection to highlight the evolution of research trends in these areas over time. A comprehensive summary of the reviewed works, including methodology, key findings, and limitations, is provided in Table 1 of Appendix 6.

In selecting papers for this study, we adopted a structured search strategy across Google Scholar, ResearchGate, ScienceDirect, and Springer. The primary search terms included "workplace injuries," "return-to-work statistics," "workers' compensation after workplace injury," and "occupational injuries return time." Additionally, we incorporated terms within logical connectives such as ("workplace injury" OR "occupational injury"

OR "work-related injury") AND ("machine learning" OR "artificial intelligence" OR "deep learning" OR "predictive analytics"). The inclusion criteria restricted the selection of articles to those published in English between January 2015 and July 2024 that applied machine learning or predictive analytics to workplace injuries. Studies focused solely on clinical rehabilitation without ML or advanced analytics were excluded. The timeframe was selected because machine learning applications in occupational health began to gain prominence around 2015, largely due to the increased digitalization of occupational data.

Risk prediction

Machine learning and predictive models

(A) *Prediction of Return-to-Original-Work After an Industrial Accident Using Machine Learning and Comparison of Techniques (2018)* [5] In their study, Lee and Kim investigate the use of machine learning techniques to predict return to original work (RTOW) after an industrial accident. They analyze data from the fourth Panel Study of Worker's Compensation Insurance (PSWCI), which sampled 2,000 of 89,921 patients who completed their recovery period in 2012. The dataset was divided into training and test subsets to develop and validate the predictive models. The researchers applied advanced machine learning techniques, including logistic regression, decision trees, random forests [24], and support vector machines (SVM) [25], to predict RTOW. They assessed the models' performance using accuracy, sensitivity, specificity, kappa value, and the area under the curve (AUC). Among the models, the random forest showed the best accuracy (78.75%) and specificity (91.38%), while the decision tree model had the highest sensitivity (67.34%) and kappa value (0.4431). Key predictors identified for RTOW included earned income, working status, satisfaction with employer, and maintenance of a relationship with the company. The random forest model identified the maintenance of the relationship with the company as the most significant variable, while the decision tree and SVM models highlighted working status as the most important factor. The logistic regression model emphasized earned income as an important predictor. The study concludes that machine learning techniques can predict RTOW, with the random forest model performing best. However, the differences in performance among the models were minor. The authors recommend using models like random forests for their accuracy and decision trees for their interpretability and practical application [26]. This finding is consistent with previous research indicating that employee characteristics significantly influence RTOW [27]. Additionally, the study highlights that machine learning techniques offer more flexible and robust predictive capabilities compared to traditional statistical methods [28]. The authors underscore the importance of company-related factors in facilitating successful return to work after industrial accidents, in line with earlier findings that emphasize the role of workplace support and employment conditions in return-to-work outcomes [29]. Lee and Kim employ a limited set of predictive variables, such as earned income and company-related factors that overlook important influences on RTW outcomes, like mental health and social support. Additionally, the research employs

simple machine learning models and relies on easy hyperparameter tuning, missing the opportunity to use more advanced techniques such as Bayesian optimization, to find the optimal set of hyperparameters for a machine learning model, which could enhance predictive performance.

(B) *A Machine Learning-Based Predictive Model of Return to Work After Sick Leave (2019)* [30] In their research, Na and Kim develop a predictive model for RTW after sick leave using a machine-learning algorithm. Utilizing panel data from 2,000 participants collected by the Labor Welfare Research Institute of the Korea Workers' Compensation & Welfare Service, the authors employed a Gradient Boosting Machine (GBM) to build their model. The data included 1,686 males and 314 females, focusing on individual, work-related, injury-related, and psychological factors. The GBM model showed excellent performance in binary classification (returned to work vs not working) with an F-measure of 0.839 and an Area Under the Curve (AUC) of 0.942, as detailed in the results table (Fig. 1). However, the model's performance for a three-group classification (return to original workplace, return to other workplaces or self-employed, and not working) was suboptimal. The study highlights the importance of various factors influencing RTW, such as perceived work ability, self-esteem, and the use of rehabilitation services. To address missing data, Multiple Imputation by Chained Equations (MICE) was utilized [31], while the Synthetic Minority Oversampling Technique (SMOTE) was applied to manage the dataset's imbalance [32]. Additionally, the study acknowledges the importance of psychological factors such as self-esteem and self-efficacy [33, 34]. The paper emphasizes the potential of machine-learning algorithms in accurately predicting RTW, although it notes the limitations in differentiating the type of RTW. In addition to the limitations noted by the authors, the study could benefit from exploring other machine learning models such as XGBoost or Neural Networks, which might improve predictive accuracy. Advanced hyperparameter optimization techniques, like Bayesian Optimization, could further refine the model's performance. Moreover, the use of Gradient Boosting Machine (GBM), while powerful, lacks interpretability. Applying explainability techniques like SHapley Additive exPlanations (SHAP) could make the model's predictions more transparent and actionable by understanding the contribution of each predictor [35, 36], thereby enhancing its practical utility in workplace settings. These enhancements could address the study's current gaps and provide more robust predictive insights.

(C) *Workplace Incident and Injuries Prevention Using Machine Learning (2023)* [37] In their paper, the authors present a comprehensive approach to preventing workplace incidents and injuries using machine learning techniques. They address the

	F-Measure	AUC	Accuracy	Sensitivity	Specificity	Positive Predictive Value	Negative Predictive Value	False-Positive Rate	False Discovery Rate	False-Negative Rate
Binary classification	0.839	0.942	0.907	0.830	0.939	0.849	0.930	0.062	0.151	0.170
Three-group classification	—	—	0.602	—	—	—	—	—	—	—
Return to the original workplace	0.731	—	—	0.803	0.777	0.658	0.881	—	—	—
Other workplace or self-employed	0.526	—	—	0.571	0.741	0.548	0.759	—	—	—
Not working	0.484	—	—	0.405	0.879	0.585	0.777	—	—	—

Fig. 1 Performance Metrics of the Binary and Three-Group Classifications

challenges associated with imbalanced data distributions and overlapping characteristics between incident and normal cases by employing advanced anomaly detection and classification algorithms. The study utilizes a dataset of 8.93 million records of hourly employee timecards, enhanced with project descriptions, customer details, and workplace conditions, collected from August 2017 to May 2019. Approximately 15,000 records were labeled as incidents. The authors applied several machine learning methods, including Isolation Forest, Autoencoders, and Positive-Unlabeled (PU) classification, to identify incidents. Isolation Forest was used for its effectiveness in anomaly detection, while Autoencoders provided unsupervised learning capabilities for identifying unusual patterns [38]. PU classification was employed to predict future incidents based on employee concern reports. To improve prediction accuracy, the authors used various data preprocessing techniques and engineered features to capture seasonality and other relevant patterns. They utilized H2O's AutoML framework for classification tasks, balancing the dataset using H2O's "balanced_classes" function [39]. The study demonstrated that the proposed models could effectively predict incidents, with the PU classification model achieving an 80% accuracy rate. This approach improves workplace safety by enabling early detection of potential incidents, allowing for timely intervention. The results highlight the potential of machine learning to enhance occupational health and safety through predictive analytics and anomaly detection. The paper reveals limitations in addressing the severe data imbalance, as it primarily relies on basic under-sampling and AutoML's "balance_classes" function. These methods may not fully address the issue due to potential loss of valuable information, sensitivity to noise, and the absence of more advanced techniques. For instance, the authors could have used techniques such as SMOTE (Synthetic Minority Over-sampling Technique) or Balanced Random Forest, which are designed to better manage class imbalance and capture complex patterns in the data.

Psychosocial and behavioral risk factors

(A) *Predicting Return to Work in a Heterogeneous Sample of Recently Injured Workers Using the Brief ÖMPSQ-SF (2019)* [40] In this research, the authors evaluate the predictive validity of the Örebro Musculoskeletal Pain Screening Questionnaire-short version (ÖMPSQ-SF) for determining the time to return to pre-injury work duties (PID), which is equivalent to RTW, among a diverse sample of workers with recent work-related soft tissue injuries. The study also examines the appropriateness of a cut-off score of 50/100 for injury case identification. The participants, consisting of injured workers from six public hospitals in Sydney, Australia, were recruited by insurance case managers within 5–15 days of their injury. The study involved administering the ÖMPSQ-SF via telephone by the case manager, followed by statistical analysis to predict days to return to PID. Cox proportional hazards regression analysis was employed to determine the predictive power of the ÖMPSQ-SF total score on the number of days to return to PID. The results indicated that for every 1-point increase in the ÖMPSQ-SF score, the predicted chance of returning to work reduced

by 4% ($hazard_ratio = 0.96$, $p < 0.001$). This demonstrates the tool's effectiveness in predicting delayed return to work. Receiver operator characteristic (ROC) analysis was used to determine the optimal cut-off score for predicting return to PID within 2–7 weeks. The analysis suggested that a score of 48 was optimal for sensitivity (0.65) and specificity (0.79), but the authors recommended using a pragmatic cut-off of 50/100 for future studies, aligning with previous findings by Linton et al. [41]. The findings underscore the utility of the ÖMPSQ-SF as an efficient screening tool for early identification of workers at risk of delayed RTW. This is particularly significant given the association between delayed RTW and persistent health problems, loss of employment, and substantial costs for individuals, families, employers, and communities [42, 43]. The ability to predict RTW using psychosocial factors, such as pain severity, anxiety, and depression, is crucial since these factors are potentially modifiable through appropriate interventions [44]. The study's methodological strengths include its prospective design, the use of an objective outcome measure (insurance company documented RTW), and the inclusion of a heterogeneous sample of injured workers, enhancing the generalizability of the findings. The authors highlight the importance of early intervention based on psychosocial risk factors to improve RTW outcomes and reduce the likelihood of long-term disability.

Return-to-work strategies

Rehabilitation and occupational therapy

(A) *Predicting Return to Work After Traumatic Brain Injury Using Machine Learning and Administrative Data (2023)* [45] The authors investigate the use of machine learning to predict RTW after traumatic brain injury (TBI) using a large, population-based dataset. The dataset combines discharge, claims, and social security data from patients hospitalized with TBI in Belgium in 2016. The study aims to determine whether administrative data can accurately predict RTW one year after TBI and to evaluate the impact of distinguishing between mild and moderate-to-severe TBI on model performance. The researchers employed three machine learning algorithms: elastic net, logistic regression, random forest, and gradient boosting, to predict RTW. The performance of these models was evaluated using metrics such as accuracy, sensitivity, specificity, and the area under the curve (AUC). For binary classification, the Random Forest model achieved the highest accuracy of 83.0% and a receiver operating characteristic curve (ROC) AUC of 84.6% in the full study population (Fig. 2). Sensitivity was highest in the Gradient Boosting model for mild TBI, reaching 57.7%, while specificity was highest at 95.4% for moderate-to-severe TBI using Gradient Boosting. For multiclass classification Random Forest again showed the highest accuracy in the full study population, with 75.8% and a ROC AUC of 82.3%. The highest sensitivity in multiclass classification was 55.1%, achieved by the Random Forest model for the full study population, while specificity was highest at 91.8% in the same model. These results indicate that while Random Forest generally outperformed other models in terms of accuracy and ROC AUC, the Gradient Boosting model provided higher sensitivity and specificity in certain TBI subgroups,

	Binary Accuracy	Sensitivity	Specificity	ROC AUC	Multiclass Accuracy	Sensitivity	Specificity	ROC AUC
Mild TBI								
Elastic Net Regression	82.9%	54.6%	94.1%	85.2%	75.5%	49.2%	90.6%	81.3%
Random Forest	82.8%	54.6%	91.5%	84.6%	75.6%	50.0%	91.0%	84.0%
Gradient Boosting	82.6%	57.7%	91.9%	85.3%	74.7%	48.4%	90.5%	82.4%
Moderate-to-severe TBI								
Elastic Net Regression	83.7%	56.6%	95.2%	85.0%	72.1%	53.1%	90.9%	82.2%
Random Forest	81.8%	56.9%	94.9%	82.6%	71.7%	52.7%	90.8%	84.4%
Gradient Boosting	82.9%	55.1%	95.4%	83.0%	70.9%	51.5%	90.6%	85.6%
Full study population								
Elastic Net Regression	82.9%	56.6%	95.2%	85.2%	73.9%	51.2%	91.1%	83.5%
Random Forest	83.0%	56.6%	95.4%	84.6%	75.8%	55.1%	91.8%	82.3%
Gradient Boosting	81.6%	55.8%	93.5%	84.7%	74.5%	52.0%	91.3%	82.5%

ROC AUC = Area Under the Receiver Operator Curve.

Fig. 2 Predictive performance on hold-out test dataset

highlighting the complexity and variability of predicting RTW outcomes based on TBI severity. Notable predictors for RTW included pre-injury employment status, type of employment (blue-collar vs. white-collar), and sector (public vs. private). Socioeconomic status (SES), age, and acute length of stay (LOS) were also significant predictors. The study found that modeling mild and moderate-to-severe TBI separately did not result in differences in model performance or feature importance. This study builds on previous research by Maas et al. [46], which emphasized the need for integrated approaches to improve prevention, clinical care, and research in TBI. The authors find that while administrative data is useful for predicting RTW, more data sources are needed to improve the accuracy of patient-specific predictions. This could include clinical, functional, and environmental data to capture the complex interaction of factors influencing RTW after TBI. The findings align with Stromberg et al. [47], who used decision tree methodology to predict RTW after moderate to severe TBI and found similar complexities in modeling RTW outcomes.

Employer-driven interventions

(A) *Predicting Return to Work After Cardiac Rehabilitation Using Machine Learning Models (2023)* [48] In their study, Yuan et al. explore the potential of using machine learning models to predict RTW after cardiac rehabilitation (CR). The researchers focus on patients admitted to the University of Malaya Medical Centre due to cardiac events and utilize data collected between 2015 and 2019. The dataset comprises 929 patient records, which include variables such as sociodemographic details, medical history, mental health status, and CR program specifics. The study evaluates eight different machine learning models: Decision Tree, Random Forest, AdaBoost, XGBoost, CatBoost, Support Vector Machine (SVM), Complement Naive Bayes, and Multiple Logistic Regression. Significant features were identified using multiple methods, including Recursive Feature Elimination (RFE) and model-based feature importance, to ensure robustness given the complexity of the dataset. While logistic regression was initially considered, its limitations led to a focus on more suitable feature selection techniques such as RFE, which aligned better with the non-linear nature of the data. The AdaBoost model, when applied to the top 20 features selected through RFE, achieved the highest performance with an area under the curve (AUC) score of 92.4% (Fig. 3). This performance was supe-

rior compared to other models and feature sets, demonstrating the efficacy of AdaBoost in handling the dataset's complexity and predicting RTW outcomes. The results underscore the importance of several key predictors for RTW, including age, duration of CR, exercise intensity during CR, and mental health indicators such as depression scores. Notably, the study finds that patients who returned to driving after CR had a significantly higher likelihood of RTW, highlighting the role of functional recovery in determining work readiness. The authors emphasize the economic and health-related benefits of successful RTW for cardiac patients. RTW not only improves health-related quality of life (HRQoL) and reduces depression and anxiety but also reduces financial problems associated with prolonged unemployment [49]. The study notes that in Malaysia, the economic impact of cardiovascular disease-related productivity loss is substantial, amounting to RM 2.7 billion ($595 million US dollars). The dataset's origin from a single medical center limits the generalizability of the findings. Including data from multiple centers across different regions could improve the robustness of the models. Moreover, the absence of key work-related variables, such as job satisfaction and physical demands, may have affected predictive accuracy. Exploring alternative feature selection techniques could further optimize model performance. The study's focus on standard metrics like AUC and accuracy might have overlooked other critical measures like precision or area under the precision recall curve (AUPRC), which are important in imbalanced datasets [50, 51]. Finally, the study's applicability may be limited by the specific economic and cultural context of Malaysia, suggesting the need for broader validation in different settings.

(B) *Workplace Factors Associated with Return to Work After Mild-to-Moderate Traumatic Brain Injury (2023)* [52] This work by Fure et al. explores work-related predictors of RTW 6 and 12 months after mild-to-moderate traumatic brain injury

Classifiers	Number of features	Accuracy	ROC AUC	Sensitivity	Specificity
AdaBoost	**10**	**0.842 (±0.080)**	**0.913 (±0.065)**	**0.911 (±0.060)**	**0.700 (±0.153)**
SVM	10	0.825 (±0.064)	0.868 (±0.071)	0.903 (±0.052)	0.667 (±0.176)
CatBoost	10	0.815 (±0.079)	0.862 (±0.063)	0.903 (±0.083)	0.633 (±0.172)
XGBoost	10	0.771 (±0.095)	0.850 (±0.072)	0.871 (±0.096)	0.567 (±0.225)
Random Forest	10	0.766 (±0.074)	0.805 (±0.086)	0.887 (±0.086)	0.517 (±0.146)
Logistic regression	10	0.765 (±0.104)	0.795 (±0.096)	0.886 (±0.070)	0.517 (±0.214)
Decision Tree	10	0.689 (±0.086)	0.640 (±0.093)	0.781 (±0.098)	0.500 (±0.157)
Complement Naive Bayes	10	0.359 (±0.055)	0.463 (±0.057)	0.049 (±0.080)	1.000 (±0.000)
AdaBoost	**20**	**0.864 (±0.084)**	**0.924 (±0.067)**	**0.928 (±0.047)**	**0.733 (±0.196)**
SVM	**20**	**0.853 (±0.072)**	**0.901 (±0.071)**	**0.919 (±0.077)**	**0.717 (±0.209)**
XGBoost	20	0.799 (±0.087)	0.863 (±0.086)	0.887 (±0.088)	0.617 (±0.137)
CatBoost	20	0.820 (±0.070)	0.862 (±0.069)	0.918 (±0.087)	0.617 (±0.193)
Random Forest	20	0.797 (±0.091)	0.846 (±0.071)	0.910 (±0.073)	0.567 (±0.211)
Logistic regression	20	0.765 (±0.110)	0.842 (±0.088)	0.837 (±0.124)	0.617 (±0.209)
Decision Tree	20	0.711 (±0.103)	0.669 (±0.099)	0.788 (±0.148)	0.550 (±0.177)
Complement Naive Bayes	20	0.380 (±0.040)	0.460 (±0.062)	0.128 (±0.095)	**0.900 (±0.117)**
AdaBoost	30	0.820 (±0.079)	0.900 (±0.070)	0.894 (±0.097)	0.667 (±0.111)
SVM	**30**	**0.857 (±0.071)**	**0.898 (±0.076)**	**0.910 (±0.048)**	**0.750 (±0.162)**
CatBoost	30	0.814 (±0.079)	0.886 (±0.052)	0.917 (±0.088)	0.600 (±0.225)
XGBoost	30	0.820 (±0.083)	0.873 (±0.071)	0.902 (±0.085)	0.650 (±0.146)
Random Forest	30	0.798 (±0.065)	0.854 (±0.082)	0.926 (±0.073)	0.533 (±0.153)
Logistic regression	30	0.782 (±0.087)	0.825 (±0.083)	0.879 (±0.084)	0.583 (±0.239)
Decision Tree	30	0.695 (±0.087)	0.640 (±0.087)	0.796 (±0.149)	0.483 (±0.200)
Complement Naive Bayes	30	0.482 (±0.115)	0.571 (±0.186)	0.328 (±0.124)	**0.800 (±0.153)**
AdaBoost	40	0.831 (±0.086)	0.897 (±0.074)	0.902 (±0.095)	0.683 (±0.123)
SVM	**40**	**0.852 (±0.060)**	**0.894 (±0.054)**	**0.902 (±0.053)**	**0.750 (±0.142)**
CatBoost	40	0.814 (±0.080)	0.875 (±0.052)	0.918 (±0.095)	0.600 (±0.179)
Random Forest	40	0.820 (±0.047)	0.874 (±0.073)	**0.943 (±0.068)**	0.567 (±0.179)
XGBoost	40	0.798 (±0.094)	0.862 (±0.066)	0.895 (±0.093)	0.600 (±0.179)
Logistic regression	40	0.744 (±0.078)	0.798 (±0.063)	0.806 (±0.057)	0.617 (±0.193)
Decision Tree	40	0.722 (±0.070)	0.664 (±0.076)	0.829 (±0.127)	0.500 (±0.192)
Complement Naive Bayes	40	0.590 (±0.140)	0.659 (±0.159)	0.479 (±0.192)	**0.817 (±0.146)**

Bolded text show high performance obtained.

ROC: receiver operating characteristic; AUC: area under curve; SVM: Support Vector Machine; MLR: multiple logistic regression.

Fig. 3 Performance models with features from recursive feature elimination (RFE)

(TBI). The main objective is to fill the knowledge gap regarding workplace factors influencing RTW, beyond the well-studied sociodemographic and injury-related predictors. Data were collected from patients treated at a specialized TBI rehabilitation outpatient clinic at Oslo University Hospital. The study involved 113 participants who were employed at least 50% at the time of injury and had suffered a mild-to-moderate TBI 8 to 12 weeks prior. The researchers used a multivariable linear regression model to identify significant work-related predictors while controlling for sociodemographic and injury-related factors. The primary outcome measure was work participation 12 months post-injury. The best-fitting model explained 25% of the variance in work participation at 1 year. Significant predictors included workplace predictability, quantitative demands, and rewards, as well as private or public employment, symptom burden at baseline, and sex. The study found that employees in the public sector had higher work participation rates than those in the private sector, likely due to better job security and supportive workplace policies in public enterprises [53]. Predictability in the workplace was positively associated with higher work participation, suggesting that a stable and predictable work environment helps employees with TBI manage their symptoms and perform their tasks effectively. This matches earlier research showing that predictability lowers absence days [54]. Similarly, quantitative demands were found to enhance work participation, possibly because a challenging workload may serve as a motivator and increase work-related well-being [55]. Moreover, higher rewards were associated with lower RTW, contrary to some prior findings [56]. This could reflect that employees who feel valued and recognized might not feel pressured to return before they are ready. The study highlights how complicated returning to work can be and the need for more research on workplace factors that can be modified to improve RTW outcomes. Sex and symptom burden were important predictors, with women working less than men and higher symptom burdens causing reduced work participation. This agrees with previous studies showing that severe symptoms affect returning to work[57]. The authors highlight limitations due to the small sample size (113 participants) which, when using too many predictors, risks overfitting. Furthermore, using a small sample may not fully capture the complex factors influencing the research outcomes, potentially undermining the robustness of the findings. A larger sample could enhance accuracy and comprehensiveness.

Demographic analysis

Age-related differences

(A) *A Longitudinal Study of Work-Related Injuries: Comparisons of Health and Work-Related Consequences Between Injured and Uninjured Aging United States Adults (2018)* [58] In their paper, Baidwan et al. investigate the association between physical work-related factors and injuries among aging U.S. workers, comparing the health and work-related consequences between injured and uninjured individuals. The study utilized data from the Health and Retirement Study (HRS), a nationally representative panel study of U.S. adults aged 50 and above, collected from 2004

to 2014. The cohort consisted of 7,212 workers aged 50 years and older. Physical work-requirement factors such as excessive physical effort, lifting heavy loads, and stooping/kneeling/crouching were identified as significant predictors of work-related injuries. In this work, the authors found that workers with jobs requiring these physical efforts "all or almost all the time" had a risk of injury more than two times greater than those without such requirements [58]. This finding aligns with previous research indicating that physical demands are closely linked to injury risk [59, 60]. Moreover, Baidwan et al. explored the long-term consequences of these injuries. Injured workers were found to have higher risks of functional limitations and reduced working hours over time compared to their uninjured counterparts. For instance, injured workers were almost twice as likely to report difficulties with activities of daily living, such as bathing, dressing, and walking, highlighting the significant impact of work-related injuries on personal health and daily functionality [61]. The analysis employed by Baidwan et al. generalized estimating equations (GEEs) and Cox proportional hazards models to estimate incident rate ratios (IRRs) and hazard ratios (HRs) for work-related injuries and their consequences. The results showed that workers exposed to high physical demands had significantly higher IRRs and HRs for both the occurrence and number of injuries, reinforcing the need for workplace interventions to mitigate these risks [62, 63]. Interestingly, this research also found that workers aged 55–64 years had the highest incidence rate of days-away-from-work injuries, at 115.8 cases per 10,000 full-time workers, compared to other age groups [64]. This underscores the vulnerability of aging workers to severe work-related injuries, necessitating targeted safety measures and accommodations to support this demographic [65]. The paper's strengths include its longitudinal design, large sample size, and the use of robust statistical methods to account for within-person and within-household correlations. However, it also acknowledges limitations such as the reliance on self-reported data and the potential for recall bias due to the two-year recall periods between survey waves [66].

(B) *Nonfatal Occupational Injuries to Younger Workers – United States 2012–2018 (2020)* [67] In this research, Guerin et al. analyze nonfatal occupational injuries among younger workers in the United States from 2012 to 2018, utilizing data from the National Electronic Injury Surveillance System (NEISS-Work) and the Bureau of Labor Statistics (BLS) Survey of Occupational Injuries and Illnesses (SOII). The study aims to provide a comprehensive overview of injury trends and disparities among workers aged 15–24 years compared to those aged 25–44 years, with a particular focus on understanding the factors contributing to the higher injury rates observed among younger workers. To achieve this, the researchers applied various machine learning techniques to analyze the dataset, which included information on approximately 3.2 million nonfatal job-related injuries treated in hospital emergency departments (EDs) over the seven-year period. Machine learning models, such as decision tree, random forest, and logistic regression, were employed to identify key predictors of injury, including sociodemographic factors, job characteristics, and environmental conditions. These models allowed the researchers to uncover complex patterns and relationships within the data, leading to more accurate predictions of injury risks. The analysis revealed that workers aged 18–19

years experienced the highest injury rates, with 404 injuries per 10,000 full-time equivalent (FTE) workers. This age group, along with other younger workers aged 15–24 years, consistently showed higher injury rates than their older counterparts, with annual rates ranging from 1.2 to 2.3 times higher compared to workers aged 25–44 years. One of the key findings from the machine learning models was that contact with objects and equipment was the leading cause of ED-treated injuries among all age groups, with injury rates of 182 per 10,000 FTE among workers aged 18–19 years [68]. Lacerations and punctures were identified as the most common types of injuries for workers under 25 years, while strains and sprains were more prevalent among older workers aged 25–44 years. The study also highlighted the significant role of industry type in determining injury risk. In 2018, service occupations, particularly in the leisure and hospitality industry, had the highest percentage of injuries requiring at least one day away from work among young workers. Workers aged 15–17 years in the leisure and hospitality industry, especially those in the accommodation and food services subsector, were found to have particularly high injury rates. The machine learning models underscored the importance of considering the specific characteristics of different industries and job types when assessing injury risk and designing targeted interventions. Moreover, the research emphasizes the need for targeted public health efforts to equip young workers with essential workplace safety and health competencies before they enter the workforce [69]. The study suggests that schools can play a vital role in providing foundational workplace safety education to adolescents. The National Institute for Occupational Safety and Health (NIOSH) and its partners have developed a free curriculum, Talking Safety, designed to teach young workers about job safety and health, which has been demonstrated to be effective [70]. This curriculum could be integrated into school programs to prepare young workers for the risks they may face in the workplace. Despite a general decline in overall ED-treated injury rates from 2012 to 2018, the study's findings indicate that young workers continued to experience disproportionately high rates of occupational injuries compared to adult workers. The machine learning models confirmed that younger workers are at greater risk due to factors such as lack of experience, insufficient training, and exposure to dangerous work environments [71]. The use of machine learning in this study not only provided valuable insights into the injury trends and disparities among young workers but also highlighted the potential for these techniques to inform more effective prevention strategies and policy interventions. However, the study also acknowledges the need for incorporating additional variables, such as workplace safety culture and individual worker characteristics, into future models to enhance predictive accuracy and better address the complex factors contributing to occupational injuries among young workers. The reliance on historical data also means that these models may not fully capture emerging trends or changes in workplace environments, limiting their applicability to future scenarios. Finally, the authors did not appear to address the dataset's imbalance, where injury cases are outnumbered by non-injury cases. This oversight in modeling such data could significantly undermine the accuracy of their predictions. Addressing these limitations in future

research could enhance the robustness and applicability of machine learning models in occupational health and safety.

(C) *A Detailed Analysis of Workplace Foot and Ankle Injuries (2021)* [72] In this analysis, Hoch et al. conduct a comprehensive analysis of workplace foot and ankle injuries, focusing on incidence, rates, and days lost from work across various industries, age groups, and genders. The study utilizes data from the Bureau of Labor Statistics (BLS) Workplace Injuries and Illnesses Nonfatal Cases Involving Days Away From Work: Selected Characteristics database, covering a period from 2003 to 2019 for incidence rates and from 2011 to 2019 for days lost from work. The researchers identified that, in 2018, over 96,000 foot and ankle injuries resulted in lost workdays, highlighting the significant economic burden these injuries impose on both workers and employers. The average workers compensation claim for ankle injuries in 2017–2018 was $17,210 in medical costs, with an additional $13,276 for indemnity. Similarly, foot injuries averaged $15,762 in medical costs and $11,866 in indemnity. The study revealed that the mining (44.85 foot, 17.09 ankle), transportation/warehousing (16.82 foot, 16.29 ankle), and utilities (24.41 foot, 26.69 ankle) industries had three of the five highest average median days missed from work for both foot and ankle injuries. However, only the transportation/warehousing industry was among the five industries with the highest rates of foot and ankle injuries, indicating that industries with higher days missed often did not have higher rates of injuries. A strong negative correlation ($P < 0.001$) was found between age and the rate of both foot and ankle injuries, while a strong positive correlation ($P < 0.001$) was observed between age and median days missed from work. The study calculated the incidence rates of foot and ankle injuries per 10,000 workers, which standardizes the data and accounts for differences in the number of male and female workers. This normalization allows for a more accurate comparison between genders, despite the likely higher number of male workers. Males had significantly higher rates of foot injuries (5.31 per 10,000 workers) compared to females (3.80; $P < 0.0001$), and slightly higher rates of ankle injuries (4.96 for males vs. 4.88 for females; $P = 0.453$). These findings underscore the need for targeted interventions to protect older workers and male employees, who are at higher risk for these types of injuries. The regression analysis showed a decrease in the incidence rate of ankle sprains, ankle fractures, foot sprains, and foot fractures from 2003 to 2019. This trend suggests improvements in workplace safety measures or reporting practices over the years. The study's findings are important for identifying industries that require additional safeguards to protect workers from foot and ankle injuries. The detailed analysis provided by Hoch et al. can help companies understand the industry norms for injury rates and days missed from work, enabling them to implement better safety protocols and reduce the economic impact of these injuries. However, a significant gap and limitation of this study is the absence of most machine learning techniques, though regression-based modeling is performed, which could have provided deeper insights and more predictive power regarding injury trends and

risk factors. Machine learning models such as random forests or gradient boosting could be applied to this dataset to identify more complex patterns and interactions between variables that traditional statistical methods might miss. For example, these models could help predict which workers are at the highest risk of injury based on multiple factors simultaneously, allowing for more targeted interventions. Furthermore, clustering techniques could be used to identify specific subgroups within industries or age groups that are particularly vulnerable, leading to more customized safety measures. Incorporating machine learning could thus enhance the study's findings, offering more robust and actionable recommendations for reducing workplace foot and ankle injuries.

Socioeconomics and cultural factors

(A) *Average Lost Work Productivity Due to Non-Fatal Injuries by Type in the USA (2021)* [73] In their study, Peterson et al. aim to comprehensively estimate the average lost work productivity due to non-fatal injuries in the USA by injury type. The researchers utilized data from the MarketScan medical claims and Health and Productivity Management (HPM) databases, focusing on individuals aged 18–64 with employer health insurance who were treated for non-fatal injuries in emergency departments (EDs) between October 1, 2014, and September 30, 2015. The study assessed workplace absences, short-term disability (STD), and workers' compensation (WC) absences among injury patients compared to matched controls. The analysis employed multivariable regression models to estimate the number and value of lost workdays attributable to various injury mechanisms (e.g., falls, motor vehicle accidents) and body regions (e.g., head, neck, spine). Lost workdays were valued using an average US daily market production estimate, adjusted to 2015 USD. The results revealed that the 1-year per-person average number and value of lost workdays due to all types of non-fatal injuries combined were approximately 11 days and $1,590. The range by injury mechanism varied significantly, from 1.5 days ($210) for bites and stings to 44.1 days ($6,196) for motorcycle injuries. By body region, the range was 4.0 days ($567) for other head, face, and neck injuries to 19.8 days ($2,787) for traumatic brain injuries (TBI). This study highlights the substantial economic burden of non-fatal injuries, emphasizing the importance of accurate estimates of lost work productivity to inform public health prevention activities. For instance, workplace absences due to falls were estimated at 5.3 days ($744), while motor vehicle traffic injuries resulted in an average of 23.5 days ($3,298) lost [74]. These findings are consistent with previous research on the economic impact of workplace injuries, such as the costs associated with motor vehicle accidents[75]. One notable finding is the high cost of lost work productivity associated with TBIs. This aligns with earlier studies indicating that TBIs often lead to long-term disability and significant economic costs. Additionally, the study underscores the variability in lost workdays and

costs across different injury types, reflecting the diverse nature of non-fatal injuries and their impact on productivity. Peterson et al. also point out several limitations, including the use of a convenience sample that may not be nationally representative, and the exclusion of long-term disability beyond one year. Furthermore, the study did not account for non-market productivity losses, such as household work, nor did it consider presenteeism (reduced productivity while at work) [76].

Discussion

The survey on workplace injuries provides a broad perspective on three key areas: risk prediction, RTW strategies, and demographic analysis. By examining these areas, we can identify both the advancements made and the gaps that remain in understanding and addressing workplace injuries. Machine learning has emerged as a powerful tool in this field, offering enhanced capabilities for predicting risks and improving RTW outcomes. However, the application and effectiveness of ML vary across these areas, revealing both similarities and distinct differences in research focus and methodology.

Prediction and Identification of Risk Factors for Workplace Injuries

Machine learning has significantly advanced the ability to predict and identify risk factors for workplace injuries. Several studies have demonstrated the potential of ML models to detect incidents and injuries, even in complex and imbalanced datasets. For example, Deshpande and Kumar [37] employed models such as Isolation Forest, Autoencoders, and Positive-Unlabeled (PU) classification to effectively handle unbalanced data distributions and overlapping characteristics between incident and normal cases. This approach highlights the strength of ML in anomaly detection and classification, providing an early warning system for potential workplace hazards. Similarly, Na and Kim [30] developed a predictive model for RTW after sick leave using a GBM, which underscored the importance of considering a range of individual and injury-related factors to improve prediction accuracy. Their work demonstrated that ML models could outperform traditional statistical methods, particularly in handling complex datasets where multiple variables interact in non-linear ways. Lee and Kim [5] further explored the use of machine learning in predicting RTOW after industrial accidents, comparing various ML techniques to optimize accuracy. Their study found that while random forests offered the best performance in terms of accuracy, decision trees were particularly useful for their interpretability. This comparison underscores a common theme in ML applications: the trade-off between model complexity and interpretability. More complex models like random forests or gradient boosting machines often provide better performance, but simpler models like decision trees offer greater interpretability, which can be crucial in practical applications where understanding the model's decisions is important for stakeholders.

One of the similarities across these studies is the focus on improving prediction accuracy by incorporating a wide range of variables, including sociodemographic, psychological, and work-related factors. However, a notable difference lies in the extent to which these studies explore advanced ML techniques. While some studies, like Deshpande and Kumar [37], leveraged cutting-edge anomaly detection methods such as isolation forest and autoencoders, others relied on more traditional ML approaches like logistic regression or decision trees. This highlights a gap in this domain, where, studies fail to leverage state-of-the-art machine learning techniques, potentially limiting the predictive power and insights that could be gained from using more advanced approaches.

The machine learning approaches most commonly used in current workplace injury studies include ensemble models such as Random Forests and Gradient Boosting Machines, and simpler algorithms like logistic regression, decision trees, and support vector machines (SVMs) [4, 11, 16, 24, 27]. Some studies have incorporated basic anomaly detection models (e.g., Isolation Forests) for incident prediction. However, on one [37] of the reviewed studies, summarized in Table 1, employed deep learning models or large language models (LLMs). The use of deep learning methods (such as convolutional or recurrent neural networks) and LLMs remains an underexplored future direction. These approaches could offer enhanced capabilities for analyzing complex and high-dimensional data, particularly unstructured textual data from injury reports, safety logs, or claim narratives. Furthermore, the potential of multimodal data fusion techniques by combining structured, unstructured, and real-time sensor data could be leveraged to incorporate multiple avenues of information for risk prediction. Future research should consider applying these advanced models.

Another gap is related to wearable technologies. These tools enable continuous tracking of workers' physical conditions and environmental factors, providing timely alerts to prevent accidents. The integration of wearable devices with predictive ML models can further enhance safety measures by identifying high-risk scenarios and enabling proactive interventions. However, the literature reveals a gap in the widespread adoption of such technologies, like wearables, with many studies still focusing on retrospective data analysis rather than real-time predictive analytics.

Return-to-work strategies

The research on RTW strategies highlights the importance of personalized interventions that consider both physical and psychological factors. Van Deynse et al. [45] emphasized the significance of integrating physical and psychological aspects in rehabilitation programs to ensure a successful RTW after TBI. Their use of machine learning to predict RTW outcomes underscored the potential of these models to tailor interventions to individual needs. Yuan et al. [48] applied ML models to predict RTW outcomes after cardiac rehabilitation, identifying key factors such as exercise intensity and mental health status. Their findings suggest that employer-driven interventions, including tailored support and job modifications, can significantly enhance RTW outcomes for employees recovering from cardiac events. This study is a strong example of how ML can be used to inform

and improve RTW strategies by providing insights that might not be apparent through traditional analysis.

One of the similarities across studies focusing on RTW strategies is the recognition of the importance of psychosocial factors, such as self-esteem, self-efficacy, and workplace support, in facilitating successful RTW. However, a key difference is the extent to which these factors are incorporated into ML models. Some studies, like Yuan et al. [48], effectively integrated these variables into their predictive models, while others relied more on traditional factors like injury severity and work type.

A significant gap in this area is the limited exploration of deep learning models and large language models (LLMs) for analyzing textual data from safety reports and other unstructured sources. Furthermore, the application of explainable AI methods, such as the SHapley Additive exPlanations (SHAP) framework [77], can help ensure model transparency and facilitate stakeholder acceptance in RTW decision-making. These advanced techniques could potentially unlock deeper insights from vast amounts of safety data, improving the accuracy of RTW predictions and enabling a more detailed understanding of the factors influencing RTW outcomes.

Demographic analysis and injury rates

The demographic analysis of workplace injuries reveals significant disparities in injury rates across different age groups, industries, and occupations. Guerin et al. [67], Baidwan et al. [58], and Hoch et al. [72] all contributed to understanding how demographic factors influence workplace injuries. For instance, Guerin's study on younger workers revealed unique risks faced by this demographic, emphasizing the importance of targeted strategies to protect them. One of the main similarities across these studies is the recognition of age as a significant factor influencing injury rates and outcomes. However, there are differences in the findings regarding the specific risks associated with different age groups. For example, while Baidwan et al. [58] found that older workers are more susceptible to severe injuries, Guerin et al. [67] highlighted the higher injury rates among younger workers, particularly in high-risk industries like construction and manufacturing. Another difference lies in the methodologies used to analyze demographic factors. While some studies utilized advanced statistical techniques to control for confounding variables, others relied on simpler descriptive statistics, which may not fully capture the complexities of the data. Moreover, most studies in this area did not employ machine learning techniques, which could provide more nuanced insights into the interactions between demographic factors and injury risks.

A significant gap in this area is the underutilization of ML models to analyze and predict injury risks across different demographic groups. Advanced models could be used to identify at-risk subpopulations more precisely, allowing for the development of targeted interventions that address the specific needs of these groups. In addition, the integration of socioeconomic and cultural factors into these models could further enhance their predictive power and relevance. In doing so, it is imperative that researchers employ

effective ethical frameworks and privacy-preserving methodologies to responsibly manage demographic and sensitive employee health data, ensuring compliance with regulations while providing informative demographic-specific injury prevention strategies.

Research gaps and future directions

Despite advancements in workplace injury research, several gaps and limitations remain that require further investigation. One critical area for future research is the use of more recent and reliable data. Many studies rely on outdated or limited datasets, which may not capture the current state of workplace safety or the latest industry practices. Integrating safety data with other datasets, such as health records or socioeconomic data, could provide better insights and more accurate predictions. This approach would allow for a comprehensive understanding of the factors contributing to workplace injuries and better inform targeted interventions.

In terms of trends across the three key categories (risk prediction, RTW strategies, and demographic analysis) there is an imbalance in the volume of research. Risk prediction has seen a significant number of studies, using machine learning models to identify potential workplace risks. However, RTW strategies and demographic analysis appear to be less frequently explored. This disparity suggests that while the field is heavily focused on preventing injuries, there is a relative lack of attention to the post-injury phase and the diverse impacts across different worker demographics. This gap indicates a need for more comprehensive research that balances these areas, particularly in developing personalized RTW interventions and understanding how demographic factors influence injury outcomes.

These advanced techniques could unlock deeper insights from vast amounts of safety data, improving the accuracy of risk prediction and enabling a more detailed understanding of the factors influencing RTW outcomes. The limitations of publicly available data also present a challenge. Public datasets often lack granularity, completeness, or timely updates, which can compromise the accuracy and applicability of research findings. For instance, while the Bureau of Labor Statistics (BLS) provides extensive data on workplace injuries, this data does not cover all industries equally or reflect the most current trends [78]. Furthermore, the lack of stratification by job type or occupational category in existing research limits the ability to identify sector-specific patterns in injury risk, ML feature importance, or model performance variability (e.g., construction vs. transportation). Similarly, the absence of categorization by specific workplace hazards or health conditions (e.g., musculoskeletal disorders, mental health issues) restricts condition-specific insights that could inform tailored interventions.

Researchers must consider these limitations and explore ways to supplement public data with more detailed and up-to-date sources, potentially through collaborations with industry or private sector data providers. In addition, future research should also incorporate job-specific contexts, workplace hazards, and country-specific policy considerations into injury models to produce more practical and actionable insights.

Study limitations

While this survey provides a comprehensive analysis of ML applications in workplace injury research across risk prediction, RTW strategies, and demographic insights, several important dimensions fall beyond its current scope. First, we did not stratify the reviewed literature by job type or occupational category, which could reveal sector-specific patterns in injury risk, ML feature importance, or model performance variability (e.g., construction vs. transportation). Second, although many studies included generic injury attributes, we did not categorize findings based on specific workplace hazards or health conditions (e.g., musculoskeletal disorders, mental health issues), which limits condition-specific insights. Moreover, geographical context was not analyzed in depth; while studies spanned multiple countries, our survey did not perform a comparative synthesis across foreign labor regulations, compensation systems, or different standards in data reporting that may affect ML generalizablility. Additionally, we prioritized studies that explicitly applied ML methods, excluding broader occupational health literature that could provide complementary qualitative or statistical insights. Finally, due to the novelty of LLMs and multimodal modeling in this domain, our prospective analysis on these methodologies is largely speculative and not grounded in existing evidence from the workplace injury domain.

Conclusion

Understanding workplace injuries and their impact on RTW outcomes is crucial for enhancing employee safety and well-being while minimizing economic burdens on organizations. This survey has demonstrated that accurate predictive models, particularly those using machine learning techniques, have the potential to significantly improve workplace safety and rehabilitation processes. By synthesizing the findings from various studies, we have highlighted the complexities of workplace injuries and the importance of personalized, data-driven interventions in promoting successful RTW outcomes.

The central thesis of this survey is that ML models offer superior capabilities in predicting workplace injuries and RTW outcomes compared to traditional methods. These models excel in handling complex, nonlinear relationships among multiple variables, providing more accurate and actionable insights. The ability of ML models to process and analyze large datasets, even those with imbalanced distributions, further underscores their value in occupational health and safety.

However, there are challenges associated with the implementation of ML models in this field. Data collection remains a significant problem, particularly in obtaining high-quality, comprehensive datasets that capture the full spectrum of factors influencing workplace injuries and RTW outcomes. Publicly available data often lacks the granularity and timeliness required for accurate predictions. Moreover, the integration of ML models into existing workplace safety and RTW programs requires substantial investment in technology and expertise, which may be beyond the reach of smaller organizations.

Despite these challenges, the advantages of ML models make them a powerful choice for improving workplace safety and RTW outcomes. Their ability to uncover complex patterns, predict high-risk scenarios, and enable proactive interventions provides a strong rationale for their continued development and adoption. Furthermore, as data collection methods improve and become more standardized, the accuracy and applicability of these models are likely to increase, making them indispensable tools for occupational health professionals.

Future research should focus on addressing the gaps identified in this survey, particularly in the areas of RTW strategies and demographic analysis. The development of more comprehensive datasets, which integrate health records, socioeconomic data, and real-time monitoring systems, will be crucial for advancing the accuracy and relevance of predictive models. Additionally, the exploration of advanced techniques such as deep learning models and large language models for analyzing unstructured data holds promise for discovering new insights into workplace injuries and RTW outcomes. Moreover, there is a need for more balanced research efforts across the key areas of risk prediction, RTW strategies, and demographic analysis. While risk prediction has received significant attention, the post-injury phase and the diverse impacts of workplace injuries across different demographics have been relatively underexplored. Addressing this imbalance will require a concerted effort to develop personalized RTW interventions that consider the unique needs of different worker populations and to understand how factors such as age, gender, and socioeconomic status influence injury risks and outcomes.

Finally, this survey has underscored the potential of machine learning in predicting and preventing workplace injuries and enhancing RTW outcomes. By building on the advancements made thus far and addressing the remaining gaps, future research can contribute to safer and more supportive work environments. The continued collaboration between researchers, employers, and policymakers will be key to fully using these predictive models, eventually leading to better health outcomes for workers and economic stability for organizations.

Appendix

Table 1 Summary of the surveyed papers

Paper Title	Year	Data	ML Algorithm/Analytical Method	Key Findings	Limitations
Risk Prediction: Machine Learning and Predictive Models					
Prediction of Return-to-original-work after an Industrial Accident Using Machine Learning and Comparison of Techniques	2018	Panel Survey from Labor Welfare Research Institute of the Korea Workers' Compensation & Welfare Service. 2,000 records	Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine (SVM)	Random Forest had the best overall performance for RTOW prediction; company relationship was the strongest predictor.	Limited predictor variables; did not model psychological or social factors; simple hyperparameter tuning.
A Machine Learning-Based Predictive Model of Return to Work After Sick Leave	2019	Labor Welfare Research Institute of the Korea Workers' Compensation & Welfare Service. 2,000 records	Gradient Boosting Machine (GBM)	GBM achieved high predictive performance; perceived work ability and self-esteem were critical predictors.	Limited multi-class classification accuracy; model lacked interpretability; no advanced hyperparameter optimization.
Workplace Incident and Injuries Prevention Using Machine Learning	2023	Private Data. 8.93 million records from hourly employee time cards.	Isolation Forest, Deep Learning, AutoML, and Anomaly Detection	Positive Unlabeled learning achieved almost 80% accuracy; early detection of high-risk scenarios was feasible.	Relied mainly on basic data balancing; missed advanced sampling methods like SMOTE or Balanced RF.
Risk Prediction: Psychosocial and Behavioral Risk Factors					
Predicting Return to Work in a Heterogeneous Sample of Recently Injured Workers Using the Brief ÖMPSQ-SF	2019	Data from 6 public hospitals in Sydney, Australia.	Cox Regression	ÖMPSQ-SF score effectively predicted delayed RTW; recommended pragmatic cutoff of 50.	Moderate sensitivity and specificity; sample limited to soft-tissue injuries only; no ML techniques applied.
Return to Work Strategies: Rehabilitation and Occupational Therapy					
Predicting Return to Work After Traumatic Brain Injury Using Machine Learning and Administrative Data	2023	Private Data from Belgium. 3,444 records.	Elastic Net Logistic Regression, Random Forest, and Gradient Boosting	Random Forest achieved the highest overall accuracy; pre-injury employment, sector, and LOS were key predictors.	Administrative data lacked clinical/functional measures; moderate multi-class classification performance.
Return to Work Strategies: Employer-Driven Interventions					
Predicting Return to Work After Cardiac Rehabilitation Using Machine Learning Models	2023	Data from the University of Malaya Medical Centre. 929 records.	Decision Tree, Random Forest, AdaBoost, XGBoost, CatBoost, Support Vector Machine (SVM), and Complement Naive Bayes	AdaBoost with RFE-selected features achieved AUC of 92.4%; exercise intensity and mental health were important predictors.	Single-center dataset limited generalizability; missing key occupational factors; only AUC prioritized in evaluation.
Workplace Factors Associated With Return to Work After Mild-to-Moderate Traumatic Brain Injury	2023	Data from Oslo University Hospital, Norway. 116 records.	Multivariable Linear Regression Model	Workplace predictability, public sector employment, and lower symptom burden favored RTW at 1 year.	Small sample size risks overfitting; limited external validity; no ML models applied.

Table 1 (continued)

Paper Title	Year	Data	ML Algorithm/Analytical Method	Key Findings	Limitations
Demographic Analysis: Age-Related Differences					
A Longitudinal Study of Work-Related Injuries: Comparisons of Health and Work-Related Consequences Between Injured and Uninjured Aging United States Adults	2018	Data from Health and Retirement Study (HRS) linked to Social Security Administration (SSA). 7212 records over 20,000 surveyed.	Generalized Estimating Equations (GEE), Cox Hazard Models, and Directed Acyclic Graphs (DAGs)	Injured workers faced greater risks of disability and reduced working hours; high physical demands doubled injury risk.	Self-reported data prone to recall bias; survey intervals were long (2 years).
Nonfatal Occupational Injuries to Younger Workers -United States, 2012–2018	2020	NEISS-Work Data (National Electronic Injury Surveillance System) and BLS SOII Data.	Descriptive Statistical Analysis	Younger workers (15–24 y.o.) had 1.2-2.3x higher injury rates; service sector showed highest injury rates.	Class imbalance was not addressed; predictive machine learning models were not applied; lacked control for individual-level variables.
A Detailed Analysis of Workplace Foot and Ankle Injuries	2022	Data from the Bureau of Labor Statistics (BLS) Survey of Occupational Injuries and Illnesses (SOII).	Spearman Correlation, One-Way ANOVA and Independent Samples t-Test, and Regression Analysis	Older workers lost more days despite lower injury rates; mining and transportation sectors were high-risk industries.	No ML techniques used; limited multivariate modeling of risk factors; regression models only capture linear trends.
Demographic Analysis: Socio-economics and Cultural Factors					
Average Lost Work Productivity Due to Non-Fatal Injuries by Type in the USA	2021	IBM MarketScan (Inpatient and Outpatient Services from emergency department- ED) Databases.	Multivariable Regression	Traumatic brain injuries and motor vehicle injuries led to the greatest productivity losses.	Non-representative sample; did not account for long-term disability or reduced productivity while working; findings may not extend to uninsured individuals.

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GAV carried out the concept and design of the research, performed the implementation and experimentation, and contributed to the manuscript. TMK contributed to the design of the research. TMK and RAB provided reviews to the manuscript. All authors read and approved the final manuscript.

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